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many ancillary kill mechanisms, which are capable of inflicting sufficient damage to the target even if the main beam misses the aim point. One of the ancillary kill mechanisms is the presence of a secondary cone of radiation symmetrical about the main particle beam. This radiation cone comprises X-rays, neutrons, alpha- and beta-particles etc, and is created by collision of beam particles with atoms of the air.

Another ancillary mechanism is electromagnetic pulse (EMP) generated by the presence of beam-current pulse. The EMP could play havoc with the electronic components and devices of the target. As these ancillary mechanisms are present due to interaction of charged particles, these are not present in the case of neutral-particle-beam based exoatmospheric weapons. Poor weather conditions such as presence of clouds, fog and rain severely limit the capabilities of high-energy laser weapons. Particle beam weapons have all-weather capability.

Advantages and limitations

Major advantages of particle beam weapons include speed-of-light delivery, rapid re-targeting, all-weather capability, better target interaction and shorter dwell time.

Speed-of-light delivery allows particle beam weapons to precisely target even fast-moving and evasive targets, which is a big advantage in anti-ballistic missile (ABM) defence. It also enables precise aim point selection on fast-moving targets at longer ranges.

Rapid re-targeting is enabled by use of changing magnetic field to deflect the charged-particle beam within certain limits without any requirement of physical movement. This feature is not available in neutral-particle beam weapons.

Particle beam weapons have all-weather capability and, unlike high-energy laser weapons, their use is

not limited by atmospheric factors such as rain, fog, clouds, dust, etc.

Particle beam weapons—due to their high particle energy coupled with secondary and tertiary kill mechanisms in the form of cone of radiation of X-rays, alpha- and beta-particles etc and EMP—interact much better with the target. Particle beams have much more impact damage on the target than the massless photons of the high-energy laser weapon, and the penetration increases with increase in particle energy. Presence of ancillary kill mechanisms increases manifold the probability of target damage. This makes it possible to inflict damage to the target even if the main particle beam misses the target.

Dwell-time requirement of particle beam weapons is much less as compared to that of high-energy laser weapons. In the case of charged-particle-beam based endoatmospheric weapons, due to high particle energy levels, the dwell time is almost negligible, being of the order of a few micro-seconds. In the case of neutral-particle-beam weapons for space usage, short dwell times would be required.

Major limitations of particle beam weapons include line-of-sight requirement, huge size and weight, massive power source requirement, thermal and electrostatic blooming and complicated beam control.

Particle beam weapons like high energy laser weapons are essentially line-of-sight weapons. Their efficacy is neutralised or reduced due to the presence of an object obscuring the target. Indirect fire used in artillery warfare is not feasible with line-of-sight weapons though one might think of reflectors on airborne or space-based platforms in the case of high-energy laser weapons enabling indirect fire.

Looking at the state-of-the-art in accelerator technology, the kind of size and weight requirement to